L2 · PCB DESIGN

Labels: **R** resistor · **C** capacitor · **Q** transistor · **U** IC · **J** connector.

Use **4+ layers**: signals top/bottom, power + ground inner planes. Planes = low-impedance supply/return, less over/undershoot, add C, lower L.

Bypass capacitors: a fast logic edge needs instant current to charge its load; the cap stores charge next to the IC and supplies it. About **1 per 1-2 ICs** (0.1 µF) + a bulk cap (10 µF).

Split ground planes: digital is noisy (switching spikes), analog needs a quiet ground, so separate planes joined at **ONE point** (more joins = ground loops, hum, oscillation).

Clock skew: same clock edge reaches parts at different times (path length, propagation delay, buffers) → eats timing margin.

Trace resistance

R = ρL / A A = width × thickness V = IR P = I²R

Convert to one unit first: 1 mil = 0.001 in = 0.00254 cm; 1 in = 2.54 cm.

> **EXAMPLE** 1 mil thick, 125 mil wide, 3 in long, 10 A, ρ = 1.68e-6 Ω·cm: A = 125e-6 in² = 8.06e-4 cm²; L = 7.62 cm; R = 15.9 mΩ; V = IR = 0.159 V; P = 1.59 W.

Clocks & pulses

f = 1/T. 1 ms = 1 kHz · 1 µs = 1 MHz · 1 ns = 1 GHz · 2 ns = 500 MHz.

Prefixes: T 10^{12} · G 10^{9} · M 10^{6} · k 10^{3} · m 10^{-3} · µ 10^{-6} · n 10^{-9} · p 10^{-12} · f 10^{-15}

Duty cycle = t_{high} / (t_{high} + t_{low}) × 100%. 50% means high time = low time.

Pulse width: measured at **50%** of max. Rise time: **10→90%**. Fall: 90→10%.

Overshoot above the rail, **undershoot** below ground, **ringing** = damped oscillation. Causes: poor grounding/bypassing, impedance mismatch, inductive loads, fast edges.

Propagation delay: gates need time to switch (~2 ns per foot of PCB), so everything must fit in one clock period — why we can't just raise the clock.

DFT also: modular layout following the schematic, visible labels, connectors/cooling/mounting placed for the chassis.

ADC example: 10-bit = 1 part in 1023 ≈ 4.89 mV/step; better accuracy → external ADC; small signals → op-amp gain first.

L3 · MICROPROCESSORS

µP = CPU only; needs external memory + I/O (68000, Z80, ARM, RISC-V). **µC** = µP + on-chip peripherals (ports, A/D, D/A, timers, watchdog, RTC): Arduino, Pico, TIVA.

Address bus: one-way (CPU → memory); n lines address **2^n bytes**. **Data bus:** two-way. **Control/status bus:** mixed (RESET, INTERRUPT, READ, WRITE, ADDRESS VALID); each may be in, out or both.

CISC (68000)	RISC (ARM)
many instructions & modes	few instructions
variable length (2-10 bytes)	fixed length, 1 cycle each
memory-to-memory ops	many general registers
compact code, many cycles	register-only math; LOAD / STORE
von Neumann (one memory)	Harvard OK (separate code/data)

Blocks inside a processor

State machine (microcode): decodes instructions, sequences reads/writes, why µPs are **synchronous** (clocked).

ALU: math/logic, results set the flags. **Registers:** fast local storage.

Status register (flags) · **PC** (next instruction address) · **SP** (top of stack, LIFO) · **Instruction register** (holds fetched op-code).

Interrupt controller · **Bus interface unit** (memory/IO timing) · **Debug core** · **Cache** (I and D) · **Glue logic** (NAND/NOT between blocks).

	68000	Z80
size	8-16-32 bit CISC, 1979	8-bit, 1976 (8080 code-compat.)
address	24 lines = 16 MB	16 lines = 64 KB
operand order	**source, dest**	**dest, source**
hex	$FF %bin @oct	0FFH (leading 0 if A-F)
flags	X N Z V C	S Z H P/V N C
multi-byte	big-endian	16-bit little-endian

Registers

- **68000:** D0-D7 data (byte/word/long); A0-A6 address + **A7 = SP** (user + supervisor); PC; SR (CCR flags). Address regs hold only word/long. 32-bit internal, 16-bit external bus.
- **Z80:** A accumulator (all arithmetic) + F; B C D E H L (8-bit; pairs BC DE HL = 16-bit); alternates A' F' B'...L' (EXX); IX IY index; SP; PC; I R.
- **Flags:** C carry · V overflow · Z zero · N negative · X extend (68000 only) · H half-carry, N=subtract (Z80 only).

Two's complement

- Invert all bits, add 1: $5E = 0101 1110 → 1010 0001 + 1 = **$A2**. 8-bit: −128 to +127; 16-bit: −32768 to +32767. Subtraction = add the negative. Overflow only when operands share a sign and the sum won't fit.
- **µP system parts:** core, memory interface, control interface (clock, RESET, interrupts), memory (RAM, ROM, flash, SDRAM), I/O (ports, timers, converters), glue logic. **DSP:** special signal-processing instructions.
- Notation (68000): $ hex, % binary, @ octal, none = decimal. Z80: trailing **H**, leading 0 if first digit is A-F (0FFFFH).
- 68000 has **two A7s:** user SP and supervisor SP (A7'), banked by mode. `CMP.W D0,D1` does D1−D0: Z if equal, N if D0 > D1.
- Z80 flags map: S=N, Z=Z, C=C, P/V=V (also parity). Z80-only H (half carry, BCD), N (last op was subtract).
- **Signed overflow:** V flag; both operands same sign and the true sum doesn't fit.

L4 · ASSEMBLY LANGUAGE

- The **assembler** turns mnemonics into machine code. Instruction = **operation** (verb) + **operands** (nouns). 68000: at least 16 bits (op-code word) + extension words (immediate/address, high word first). Z80: op-code byte(s) + operand bytes.
- Line format: `LABEL OPCODE OPERAND1,OPERAND2 * comment` (label optional, column 1; comment starts with * or ;).
- **Machine code:** NOP $4E71 · MOVE.B D0,D3 $1600 · ADD.L D4,D7 $DE84 · ADDI.W #$4000,D5 $0645 $4000 · Z80 ADD A,22H = C6 22.

Directive	68000	Z80
origin	ORG	ORG
equate	EQU	EQU
end of source	END	END
define byte / word	DC.B / DC.W	DEFB / DEFW
reserve space	DS.B nn / DS.W	DEFS nn / N/A
ASCII string	DC.B 'string'	DEFM 's'

Also (68000): SET = redefinable EQU · DCB = block of one value · REG = MOVEM list · MACRO · OPT. **EQU** = like #define.

- **MOVE.B D0,D1** (68000) copies D0 → D1, source unchanged; **LD B,A** (Z80) copies A → B. Size suffix **.B .W .L** = 8/16/32 bits. Z80: LD A,C · LD A,(1234H) · LD (HL),A.

> **ZERO A BLOCK** (68000) `LEA start,A0 · LEA end,A1 · loop: CMPA.L A1,A0 · BEQ done · CLR.L (A0)+ · BRA loop · done: SIMHALT`. CLR.L (A0)+ clears 4 bytes and advances A0 by 4. SIMHALT is a simulator command, not a 68000 instruction.

> **LARGEST OF FOUR BYTES** `MOVE.B (A0)+,D1 · CMP.B D1,D0 · BGE keep · MOVE.B D1,D0 · keep:` (D0 = max so far; BGE = signed). Z80: CP [HL] + JR NC (unsigned carry test), DJNZ counts.

- Line: LABEL (col 1, optional) · OPCODE · OPERANDS (comma, **no spaces**) · comment (* or ;). One instruction per line, up to 132 chars.
- Z80 assembler (Oshonsoft): pseudo-ops start with a period: .ORG .EQU .DB .DS .END. Z80 immediates: LD C,55H; LD DE,0A34BH. Absolute 8-bit only via A: LD [0FF20H],A; 16-bit LD [nn],HL stored **little-endian**.
- 68000 MOVE examples: MOVE.B D0,D1 · MOVE.W D7,$004AAA00 · MOVE.L (A3),D4.

> **ZERO A BLOCK (Z80)** no auto-increment and no 16-bit compare: `LOOP: LD A,H; CP D; JR NZ,CLR; LD A,L; CP E; JR Z,DONE; CLR: LD (HL),0; INC HL; JR LOOP`.

> **LOGIC & SHIFTS** AND/OR/EOR work on the stated size, other register bits untouched; AND #$F keeps a digit, AND #0 clears. AND.W D0,D1 with D0 = $3795AC5F, D1 = $B6D34B9D → D1 = $B6D3081D. ASL.B #3 on $01 = $08 (0 into bit 0, top bit to C and X). Counts > 8 must be in a Dn; memory shifts 1 bit, word only. **ASCII → hex:** subtract $30 (0-9), $37 (A-F), $57 (a-f), then ASL by 12, 8, 4, 0 and OR together.

ASSEMBLY GUIDE & TEMPLATES (read Z80, read + write 68000)

68000 template (EASy68K)

```
       ORG    $1000      ; code origin
START  LEA    ARRAY,A0   ; A0 = pointer
       MOVE.W #LEN,D1    ; D1 = count
       CLR.L  D0         ; sum = 0 (!)
LOOP   ADD.B  (A0)+,D0   ; use, A0 += 1
       SUBQ.W #1,D1      ; count down
       BNE    LOOP       ; until D1 = 0
       MOVE.B D0,RESULT  ; store answer
       SIMHALT           ; stop (sim only)
       ORG    $2000      ; data after code
ARRAY  DC.B   5,9,2,7    ; init bytes
LEN    EQU    4          ; constant
RESULT DS.B   1          ; reserve 1 byte
       END    START      ; entry point
```

Layout: constants (EQU) → code → SIMHALT → data (DC/DS) → END. **Write 68K in 7 steps:** (1) say it in words / flow chart; (2) assign registers (A0 pointer, D0 result, D1 count) in a header comment; (3) pick the size from the data (.B bytes, .W words/counts, .L addresses) and use it **everywhere**; (4) init pointer (LEA) and clear accumulators; (5) loop body, (An)+ walks memory by the operand size; (6) CMP then branch (signed BGT BLT BGE BLE, unsigned BHI BCS BHS); (7) store, SIMHALT, END; trace with a tiny input.

Control-flow patterns (68000)

```
* IF D0 < D1 THEN A ELSE B (signed)
       CMP.W D1,D0     ; D0 - D1
       BGE   ELSE      ; skip A if D0 ≥ D1
       ...A...
       BRA   ENDIF
ELSE   ...B...
ENDIF

* WHILE D0 ≠ 0 (test at top)
TOP    TST.W D0
       BEQ   DONE
       ...body...
       BRA   TOP
DONE
* DO...WHILE (runs ≥ 1 time)
LOOP   ...body...
       SUBQ.W #1,D0
       BNE   LOOP

* SUBROUTINE
       BSR   SUB       ; push return, jump
SUB    MOVEM.L D1-D2/A0,-(SP) ; save
       ...work...
       MOVEM.L (SP)+,D1-D2/A0 ; restore
       RTS             ; pop PC
```

Z80 template and how to read it

```
       ORG  0
START: LD   HL,TABLE ; HL = pointer
       LD   B,4      ; B = count
       LD   A,0      ; A = sum
LOOP:  ADD  A,[HL]   ; A = A + mem[HL]
       INC  HL       ; next byte
       DJNZ LOOP     ; B-1, loop if ≠ 0
       LD   [RESULT],A ; only A takes [nn]
       HALT
TABLE: DEFB 5,9,2,7
RESULT: DEFS 1
       END
```

- **Reading Z80:** (1) **dest, source** (LD B,A: A into B). (2) **[ ]** or ( ) = memory; none = register / number. (3) Only **A** does arithmetic and compares. (4) Find the loop: a label plus JR / JP / DJNZ going back up. (5) Flags come from the previous ADD / CP / DEC; JR Z / NZ / C / NC reads them. (6) BC, DE, HL are 16-bit pairs; 16 bits sit little-endian in memory. (7) **Trace**: write a small table (A, B, HL, flags) and step one line at a time. (8) 8-bit registers can't hold values > $FF: watch carry.
- **Z80 ↔ 68K:** MOVE.B D0,D1 = LD B,A · MOVE.B #5,D0 = LD A,5 · ADD.B (A0)+,D0 = ADD A,[HL] + INC HL · SUBQ + BNE = DJNZ · BEQ = JR Z · CMP.B D0,D1 = CP D (68K tests dst − src) · SIMHALT = HALT.
- **Common bugs:** size mismatch (.W on bytes) · (An)+ moves by size, not 1 · CMP operand order · signed vs unsigned branch · MOVE to An (use MOVEA) · .W/.L on an **odd** address (address error) · accumulator not cleared · forgot SIMHALT / END · MOVEM lists differ in save and restore · word counts not matching DC.W vs DC.B.

L5 · 68000 ADDRESSING, FLOW, STACK

- Subroutines: **JSR** pushes the 32-bit return address (SP − 4) and jumps; **RTS** pops it; **BSR** = relative call. Save registers: `MOVEM.L list,-(SP)` … `MOVEM.L (SP)+,list`. Z80: CALL / RET (16-bit PC).
- **Jumps:** JMP absolute; BRA relative (±32K); Z80 JR ±128, JP cc, CALL cc, RET cc (68000 has none of those

cc forms), RTE / RETI return from interrupt.

2-D ARRAY 50 × 50 longs at $4000, element (row 23, col 11): D0 = 23×50×4 = 4600 = $11F8; offset = 11×4 = 44 = $2C; `MOVE.L $2C(A0,D0),D3` with A0 = $4000.

Compare & branch

`CMP src,dst` computes **dst – src**, sets flags, stores nothing. Branch right after. **Z**=1 equal; **N**=1 result negative; **C**=carry/borrow; **V**=signed overflow; **X**=extend.

after CMP D0,D1	signed	unsigned
D1 < D0	BLT	BCS
D1 ≤ D0	BLE	BLS
D1 = D0	BEQ	BEQ
D1 ≠ D0	BNE	BNE
D1 > D0	BGT	BHI
D1 ≥ D0	BGE	BCC / BHS

Addresses are **unsigned** → use BHS/BCC. BMI: N=1. BPL: N=0. Test bit n: AND with a mask, check Z.

FOR/WHILE: test at top, BRA back at bottom. **DO-WHILE:** body first, test + branch back at bottom (runs ≥1 time). Z80: `CP r`, `DJNZ` (B–1, jump if ≠0), `BIT n,r` (Z = ~bit).

(An): big-endian, so a byte access at a long's address hits its **most**-significant byte; ADDQ.L #3 on the pointer reaches the least. abs.W / abs.L are slow and not relocatable; PC-relative is relocatable. Non-MOVE immediates have own forms (ADDI, SUBI, CMPI, ORI).

Array index: Xn = row × columns × bytes/element; d8 = column × bytes/element.

Z80 addressing: LD A,B (reg) · LD C,55H (imm) · LD B,[HL] (pair = pointer) · LD [nn],A (abs). Z80 jumps: JP cc,nn / CALL cc / RET cc exist; 68000 has none. No 16-bit CP on Z80.

if = CMP + branch around; subroutine header comment: name, purpose, registers, parameters; subroutines nest; set SP at top of RAM first.

L6 · BUSES, MEMORY CELLS, PAGING

Plain outputs can't share a bus (contention = short circuit). **Open collector (OC):** output only pulls low; a pull-up resistor makes high; fast falling edge, slow rising edge (RC): t_{rise} (10-90%) = **2.2·R·C_{total}**.

Tri-state: each output has an enable switch; one device drives (enabled), the others are **Hi-Z**.

~CE / ~CS = master control: if high, the whole chip is off (outputs Hi-Z, clocked inputs ignored). **~OE** only puts outputs in/out of Hi-Z. Chip drives the bus only when **~CE = 0 and ~OE = 0**; ~WE low = write.

Active low: ~NAME is true when low. Negative logic swaps gates (De Morgan): invert all inputs/outputs, **OR ↔ AND**, NOR ↔ NAND. NOR is not the negative-logic OR.

SRAM cell (6T): 2 cross-coupled inverters (4 T) hold the bit by feedback + 2 access transistors to BIT/~BIT. Flip-flop cell = D flip-flop + tri-state output.

Address space = 2^n bytes for n lines (capability, not installed memory). Memory is addressed in bytes. Word (16 b) on even addresses; long (32 b) on multiples of 4; double on multiples of 8. Data path width (8/16/32/64) sets performance.

Endian: big = most-significant byte at the lowest address (68000). Little = least-significant byte first (Z80 16-bit values).

Byte strobes: chip lane per byte; WR0..WR3 pick lanes: byte = 1 strobe, word = 2, long = all 4. A0 (and A1) are implied by the strobes.

Address-space sizes: 20 lines = 1 MB · 24 = 16 MB · 32 = 4 GB (2^n). Memory chip **N × w**: each chip gives w bits, a page needs bus width ÷ w chips.
68000: 32-bit internal, **16-bit external** data path; Z80 16-bit address, 8-bit data.

store	$1000	$1001	$1002	$1003
$AA55 big-endian (68000)	AA	55		
$AA55 little-endian (Z80, 16-bit)	55	AA		
$12345678 long, big-endian	12	34	56	78

L7 · MEMORY DECODING & TIMING

offset bits = $\log_2$(chip depth) page bits = address lines – offset pages = $2^{\text{page bits}}$ chips/page = bus width ÷ chip width

Offset (low) bits go straight to every chip's address pins. **Page** (high) bits + address-valid go to the **decoder**, whose outputs are the active-low ~CE of each page.

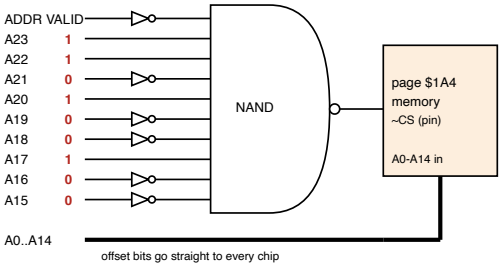
Page range: start = n × page size; end = start + size – 1. Easy way: write the page number, then offset-bit-count zeros, regroup in hex.

- **Chip select as a NAND:** output low only if every input is 1; invert each page bit that must be 0; include address-valid. Page $1A4 (A23-A15 = 1 1010 0100): inverters on A21, A19, A18, A16, A15.
- **Design steps:** (1) chips/page & page size, (2) offset vs page bits, (3) which pages exist and where, (4) decode each page uniquely.
- **74x138** (3-to-8, outputs active low): on when EN1 = 1, ~EN2 = ~EN3 = 0. ABC pick Y0-Y7. A23-A21 with ~AS on ~EN2 = 8 regions of 2 MB (16 MB map). Second 138 + one more bit = 16.
- **Memory map (68000):** vectors at $0, then ROM, peripherals, empty space, RAM at top. Paging extends a small address space: a register picks which 64 KB page is swapped in.

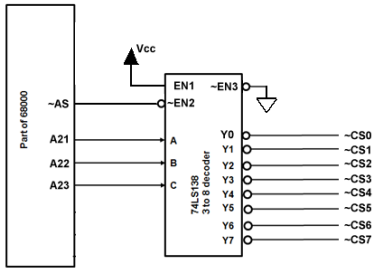
given	offset	page	pages	chips
20-bit, 8-b bus, 256K×4	A0-17	A18-19	4	8
24-bit, 8-b bus, 512K×8	A0-18	A19-23	32	32
24-bit, 8-b bus, 256K×4	A0-17	A18-23	64	128
68000, 32K pages	A0-14	A15-23	512	—

Timing & DRAM

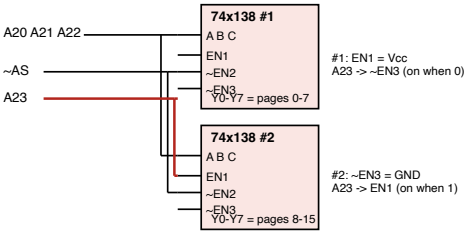
- Data valid = **decoder delay + access time** (from ~CE low). Must fit the CPU's read window, else add **wait states** (+1 clock each) or a faster part. Z80 timing is synchronous; ~WAIT stretches it.
- **Max clock, zero waits:** N·T ≥ (gates × t_{gate}) + t_{access}. HW4: 2 gates × 2 ns + 15 ns = 19 ns over 3 clocks → T = 6.33 ns = **158 MHz**.
- **SRAM:** 6T, no refresh, fast (cache). **DRAM:** 1T + C, denser, must refresh (~10-16 ms), multiplexed row/column address: 64 Mbit = 8192^2 = 13 + 13 bits. Read: row, ~RAS↓, column, ~CAS↓, data. Write: ~WE low, data before ~CAS↓. Refresh = RAS cycle per row (~2 µs each).
- **Timing method:** collect data sheets; worst-case time for address, ~OE, ~CE to reach and settle; when the CPU samples; compare incl. decoder delay; fail → wait states or faster part.
- **DMA:** peripheral asserts BUS REQUEST; CPU finishes cycle, tri-states bus, sends BUS GRANT; peripheral moves block; BUSREQ off → CPU resumes.
- **Vectors:** $000000 = initial SP, $000004 = initial PC (RESET); $000-$3FF = 256 long vectors. **Z80 paging:** sixteen 64 KB pages; value 0-15 in a register picks the one swapped in.
- **Page / chips:** page size = chips × chip size. Refresh interval = refresh time ÷ rows (16 ms / 8192 ≈ 2 µs).



Page decode (L7 ex. 3): $1A4 = 1 1010 0100. Red = required level; inverter (triangle + bubble) on every input that must be 0; ADDR VALID is active-low so it is inverted too. NAND output goes low only when all inputs are 1.



HW4 given figure: 74LS138 on the 68000, 8 pages of 2 MB (EN1 = 1, ~EN2 = ~AS, ~EN3 = 0).



HW4 #4: 16 pages, no extra gates. A20-A22 on A,B,C of both chips; ~AS on both ~EN2; A23 splits them. Pages shrink to 1 MB (20 offset bits). Single-138 version: A21-A23 on ABC, 8 × 2 MB.

68000 INSTRUCTIONS (taught)

Order: source, dest. Sizes .B .W .L (An: .W/.L only). Word/long on even addresses.

instr	does	notes
MOVE s,d	copy	N,Z set; V,C clear. Dest not An
MOVEA s,An	copy to An	.W/.L; no flags
LEA ea,An	address into An	.L; no flags
CLR ea	set to 0	Z=1, N=V=C=0
ADD/SUB s,d	d = d ± s	one side Dn; all flags (X N Z V C)
ADDI/SUBI #n	immediate	any ea but An
ADDQ/SUBQ #1-8	quick	only 1–8
ADDX	add + X flag	multi-precision
CMP s,Dn	Dn – s, flags only	CMPA (.W/.L), CMPI
AND OR EOR (+I)	bitwise	N,Z set; V,C clear. AND #$F keeps digit
NOT ea	invert bits	one operand
ASL/LSL #n,Dn	shift left	0 into bit 0, top bit to C and X; n 1–8 or Dn
EXG / SWAP	swap regs / halves	EXG .L; SWAP .W
MOVEM	save/restore list	.W/.L; -(SP) / (SP)+
NOP	nothing	$4E71

Branches & calls

BRA always (relative) · JMP abs · BEQ BNE · BMI (N=1) BPL (N=0) · signed BGE BGT BLT BLE · unsigned BHI BHS/BCC BCS BLS · **BSR/JSR** call · **RTS** return · **RTE** exception return. (Table of signed vs unsigned is in L5.)

Z80 instructions (taught)

No size suffix; register sets size. **dest, source**; [ ] = memory.

LD d,s	copy, no flags; only A uses [nn] for 8-bit
ADD A,n / A,r	add to accumulator
CP r	A – r, flags only
JP nn / JR e	jump (JR ±128); cc = Z NZ C NC
DJNZ e	B–1, jump if ≠0
BIT n,r	Z = inverse of bit n
CALL / RET / RETI	call / return / int. return
PUSH / POP rr	16-bit pairs only
EXX / EX AF,AF'	swap with alternate regs

Flags

- **X** extend (copy of C, used by ADDX) · **N** negative (top bit) · **Z** zero · **V** signed overflow · **C** unsigned carry/borrow.
- Carry = unsigned result out of range; overflow = signed result out of range. Z80: S Z H P/V N C.

COMMON TRAPS

- .W on longword data adds only the low word — match the size to the data.
- `CMP src,dst` = dst – src; signed (BGT/BLT) vs unsigned (BHI/BCS).
- (An)+ moves by operand size (1/2/4), not always 1.
- Simulator shows bytes: word $4515 = $45 then $15.
- Pages use chip **depth**; chips per page use chip **width**.
- Wait-state math: use the stated clocks per access (3), not 1.